



Supplementary Materials for

Reducing the stochasticity of crystal nucleation to enable subnanosecond memory writing

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Materials and Methods

1. Film deposition and characterization

The $\text{Sc}_{0.2}\text{Sb}_2\text{Te}_3$ (SST) and $\text{Ti}_{0.4}\text{Sb}_2\text{Te}_3$ (TST) films were deposited by co-sputtering of pure Sc and Sb_2Te_3 targets, and pure Ti and Sb_2Te_3 targets, respectively. $\text{Ge}_2\text{Sb}_2\text{Te}_5$ (GST) and Sb_2Te_3 films with the same thickness were deposited by sputtering respective compound target. The compositions of all films were measured by X-ray fluorescence spectroscopy using a Rigaku RIX 2100 system. The temperature-dependent sheet resistance changing trends of SST and GST films with the same 300 nm thickness were studied using a Linkam LMP 95 hot stage. For transmission electron microscope (TEM) samples, ~20 nm-thick SST film was deposited on supporting grid coated with ultra-thin carbon film. Amorphous SiO_2 film (~5 nm-thick), as the anti-oxidation layer, was successively deposited on top of the SST film. It was then annealed at 270 °C in vacuum for 2 min to obtain the rocksalt phase. The microstructures of the SST film were examined using bright-field TEM, selected area electron diffraction (SAED), and high resolution TEM in a JEOL 2100F TEM under 200 kV. The in situ heating crystallization of ~20 nm-thick amorphous SST film with ~5 nm-thick amorphous SiO_2 capping layer was carried out in a heating holder (Gatan 628) at heating rate of 10 °C/min. During the whole heating process, electron beam was shut off to avoid irradiation effects and only turned on for capturing images.

2. Device fabrication and electrical characterization

T-shaped phase-change random-access memory (PCRAM) devices with tungsten plug bottom electrode contact (BEC, diameter = 190 nm) were fabricated using the 0.13 μm node complementary metal-oxide semiconductor technology. In all the PCRAM devices, the thickness of the phase change material film is around 150 nm. TiN (15-nm in thickness) and Al (300-nm) films were used as the top electrode for all devices. The electrical measurements were performed using the Keithley 2600C source meter (measuring device resistance), the Tektronix AWG5002B pulse generator (generating nanoseconds voltage pulse with a minimum width of ~6 ns), and the Picosecond Pulse Labs Model 10070A pulse generator (generating picoseconds voltage pulse with a minimum width of ~100 ps and a maximum magnitude of ~7.5 V). The testing setup is described in Fig. S5.

3. DFT simulations

Density functional theory (DFT) simulations were carried out with Vienna *ab-initio* Simulations Package (VASP) (42). The Perdew-Burke-Ernzerhof (PBE) functionals (43) and the projector augmented wave (PAW) pseudopotentials (44) were used for VASP. The employed energy cutoff for plane wave was 500 eV. Cohesive energy E_{coh} of transition metal telluride (TMT) was calculated as $E_{\text{coh}}(\text{TM-Te}) = (E_{\text{TM-Te}} - N_{\text{TM}}E_{\text{TM}} - N_{\text{Te}}E_{\text{Te}}) / (N_{\text{TM}} + N_{\text{Te}})$, where N_{TM} and N_{Te} correspond to the number of TM and Te atoms, respectively. The total energy of each element was obtained by considering the stable crystalline bulk phase of each element. Γ point was used to sample large supercell models, while a 4×4×4 Monkhorst-Pack (45) k point mesh was used to sample unit cell models. The quantum chemistry bonding analyses were performed using the scheme of the crystal

orbital Hamilton populations (COHP) method (46), which is implemented in the Local Orbital Basis Suite Towards Electronic-Structure Reconstruction (LOBSTER) code (47). The LOBSTER code retrieves the wavefunction information from the plane-wave DFT calculations and projects them onto localized atomic basis sets (48).

Rocksalt Sb_2Te_3 and Sc_2Te_3 are modeled with a $3\times 3\times 3$ cubic supercell, containing 180 atoms, while YTe, MnTe, ZnTe, CdTe, and HgTe are modeled with unit cell, containing 8 atoms. The employed lattice parameters are 6.10, 5.86, 6.18, 5.77, 5.78, 6.13, and 6.15 Å, for rocksalt Sb_2Te_3 , Sc_2Te_3 , YTe, MnTe, ZnTe, CdTe, and HgTe, close to their experimental values (49). Rocksalt SST (YST, ZnST, CdST) and GST are modeled with 180 (8 Sc/Y/Zn/Cd, 64 Sb and 108 Te) and 196 (44 Ge, 44 Sb and 108 Te) atoms in a $3\times 3\times 3$ supercell cubic box with their experimental lattice parameters, 18.30 and 18.09 Å.

4. DFMD simulations

DFT-based molecular dynamics simulations (DFMD) were performed by employing a Car-Parrinello-like scheme (50), as implemented in the CP2K package (51). The Kohn-Sham orbitals were expanded in a Gaussian-type basis set of triple-zeta plus polarization quality, whereas the charge density was expanded in plane waves, with a cutoff of 300 Ry. Goedecker (52) pseudopotentials and the Perdew-Burke-Ernzerhof (PBE) functionals (43) were used. The Brillouin zone was sampled at the Γ point of the supercell. The time step for the simulations was 2 femtoseconds.

Amorphous SST, YST and GST models ($3\times 3\times 3$ cubic supercell, containing 180 atoms) were generated following a melt-quench approach: the crystalline models were firstly heated up to 3000 K for randomization, then were quenched down to the melting temperature 1000 K and equilibrated for 30 ps, and finally were quenched down to 300 K in 80 ps. These amorphous models were annealed at 600 K for 50 ps for the analysis of primitive rings. The bulk diffusion constants of GST and SST are 1.53 (Ge 2.03, Sb 1.56, Te 1.33) and 0.96 (Sc 0.39, Sb 1.24, Te 0.85), respectively (the unit is $\times 10^{-10}$ m²/s). The density was fixed at the experimental value of the crystalline phase, which is relevant to phase-change memory cells (in a confined volume). The density variation by a few percentages does not lead to any qualitative difference.

Regarding the crystallization simulations of the bigger supercell model ($4\times 4\times 4$ cubic supercell, containing 428 atoms: 20 Sc, 152 Sb and 256 Te atoms), a ScTe seed was prefixed while randomized at 3000K and equilibrated at 1000 K. Then the model was quenched down to 600 K, and the constraint was completely released at 600 K. After a short equilibration at 600 K over 30 ps, the kinetics of the model was followed for the study of crystallization.

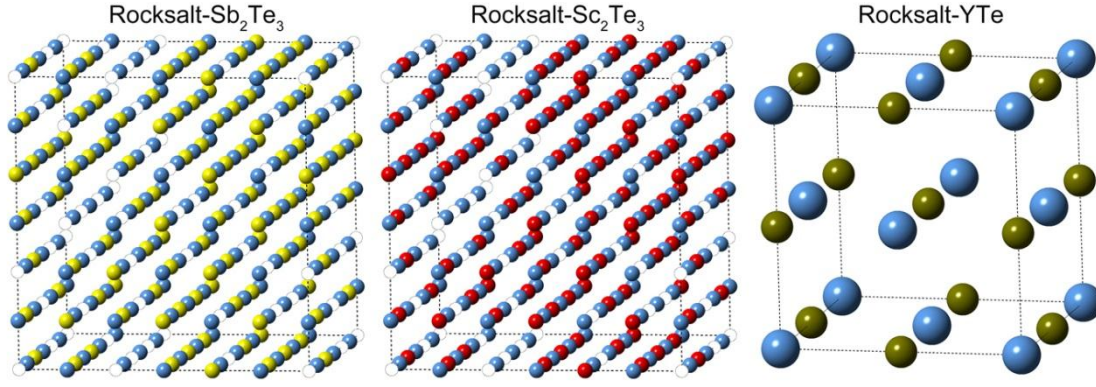


Fig. S1. Rocksalt Sb₂Te₃ and TMTs. Rocksalt Sc₂Te₃ shares similar structural feature with rocksalt Sb₂Te₃, in that atomic vacancies occupy 1/3 of the sites on the cation-like sublattice (3×3×3 supercells are constructed to comply with the exact stoichiometry), while YTe, MnTe, ZnTe, CdTe, and HgTe take the conventional rocksalt structure (for clarity, only YTe is shown here, as a representative). Atomic vacancies are rendered with hollow dashed circles, while Te, Sb, Sc and Y atoms are rendered with blue, yellow, red, and brown spheres. As discussed in the main text, the E_{coh} of rocksalt Sb₂Te₃ is -0.06 eV/atom, while that of rocksalt Sc₂Te₃, YTe, MnTe, ZnTe, CdTe, and HgTe are -1.23, -1.29, -0.05, -0.19, -0.30 and -0.03 eV/atom, respectively. Clearly, Sc₂Te₃ and YTe are considerably more robust than Sb₂Te₃. Note that all these 7 compounds are metastable crystals.

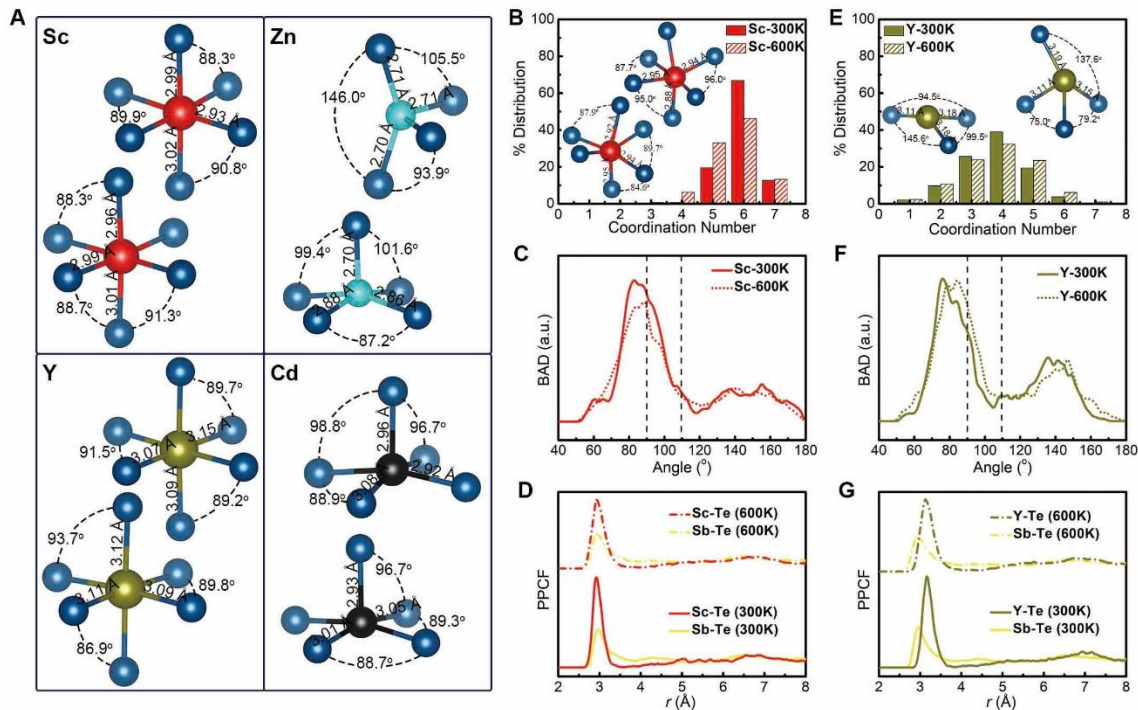


Fig. S2. (A) Representative TM atom-centered octahedral motifs in rocksalt TM-Sb₂Te₃, where TM = Sc, Zn, Y, and Cd. To match the lattice configuration of the rocksalt Sb₂Te₃, coordination number (CN) of 6, bond angle (BA) around 90°, and bond length (BL) about 3.0 Å are of necessity. Obviously, Zn- and Cd-centered motifs deviate from rocksalt Sb₂Te₃ severely, and are expected to lead to significant lattice distortions. Sc and Y elements are selected as TM dopants in the amorphous structure. Blue, red, cyan, brown, and black spheres correspond to Te, Sc, Zn, Y, and Cd atoms, respectively. (B-G) Local structural information around Sc and Y atoms in amorphous Sc and Y doped Sb₂Te₃ by DFMD simulations. Both melt-quenched amorphous phase after 30-ps equilibrium at 300 K and the phase with subsequent 15-ps post-annealing at 600 K are analyzed. (B and E) CN distribution of Sc and Y atoms, respectively. The insets show the representative motifs. CNs of Y broadly distribute around 3~5, while CNs for Sc are mainly around 6. (C and F) Bond angle distribution (BAD) around Sc and Y atoms, respectively. For both the amorphous phase and the post-annealing one, BAD peaks are at around 85~88° for Sc, while Y obviously shows much larger angle deviation. (D and G) Partial pair-correlation functions (PPCF) for Sc-Te and Y-Te pairs, respectively. Clearly, Y-Te bonds are a bit too long, whereas Sc-Te bonds fit almost perfectly with Sb-Te bonds in bond length.

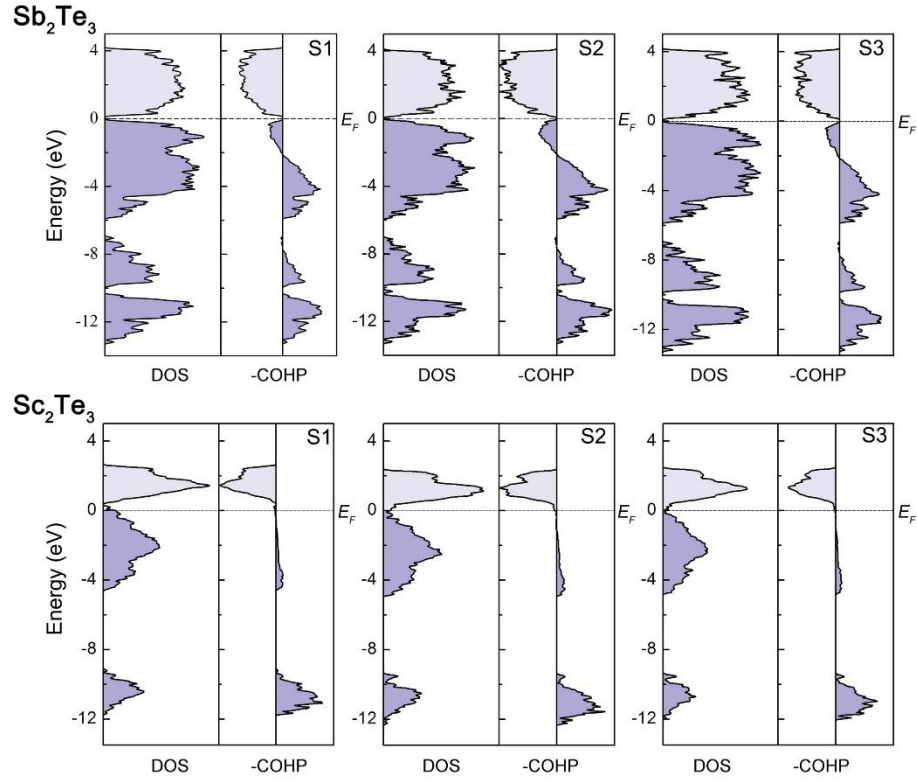


Fig. S3. Electronic density of states (DOS) and crystal orbital Hamilton populations (COHP) of rocksalt Sb_2Te_3 and Sc_2Te_3 of three independent atomic configurations (S1, S2 and S3). Atomic vacancies were randomly generated in the three models. S1 is shown in Fig. 1B.

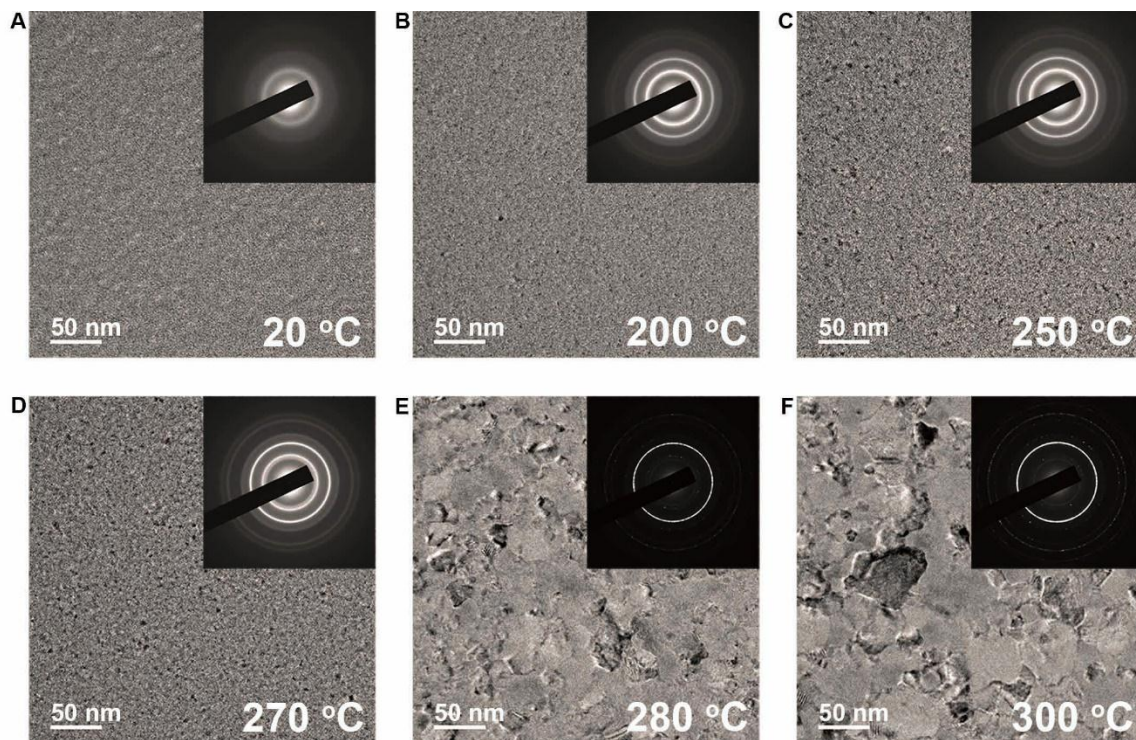


Fig. S4. In situ heating bright-field TEM images and the corresponding SAED patterns of the SST film. The images at different temperatures reveal the thermally-induced amorphous-to-rocksalt-to-hexagonal phase transformations. **(A)** The amorphous phase. **(B-D)** The rocksalt phases with smaller grains uniformly distributed over the observing area, indicating nucleation-dominated crystallization mechanism. The SEAD pattern can be indexed as rocksalt structure for which the radius squared corresponding to the different diffraction rings (1 through 8 as marked in Fig. 2C of the Main Text) in reciprocal space obeys the sequential ratio of 3 : 4 : 8 : 11 : 12 :... **(E-F)** The hexagonal phase is obtained upon further thermal annealing.

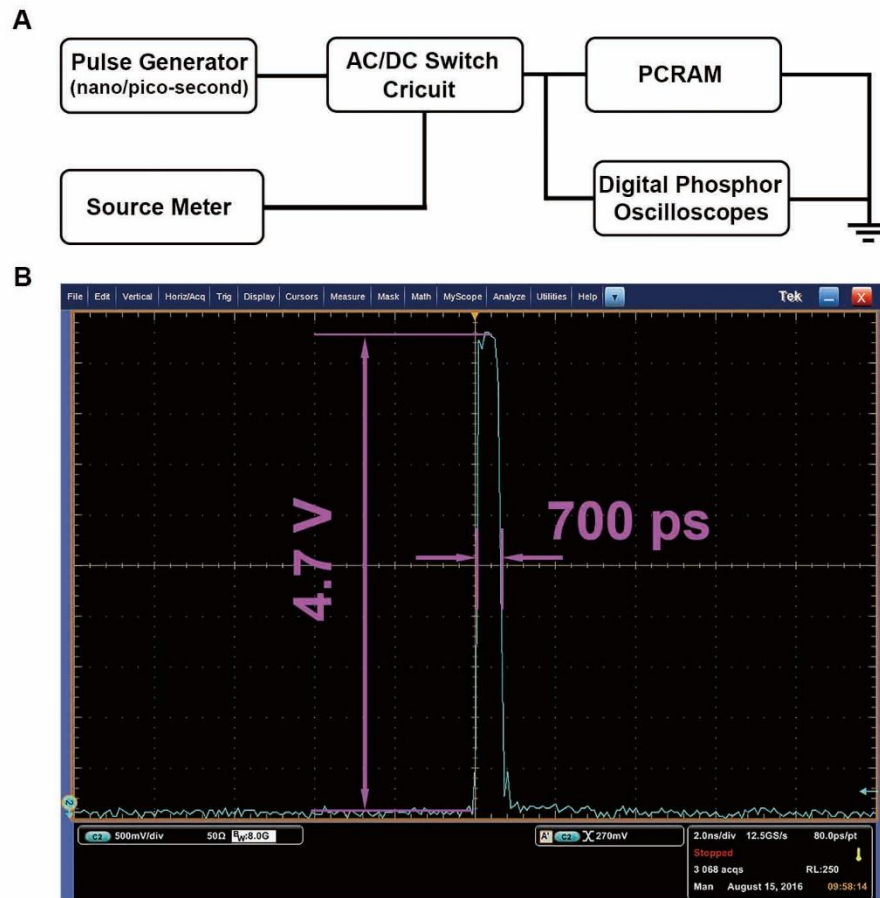


Fig. S5. Schematic of the electrical measurement setup and the waveform of the picoseconds pulse captured. **(A)** The nanosecond/picosecond pulse generator delivers voltage pulse with different widths to induce phase transformation in PCRAM device. The source meter monitors the resistance change of the device. The oscilloscope is utilized to verify the waveform applied to the device. **(B)** The captured waveform of 700-ps pulse with 4.7 V magnitude for implementing SET operation of SST device.

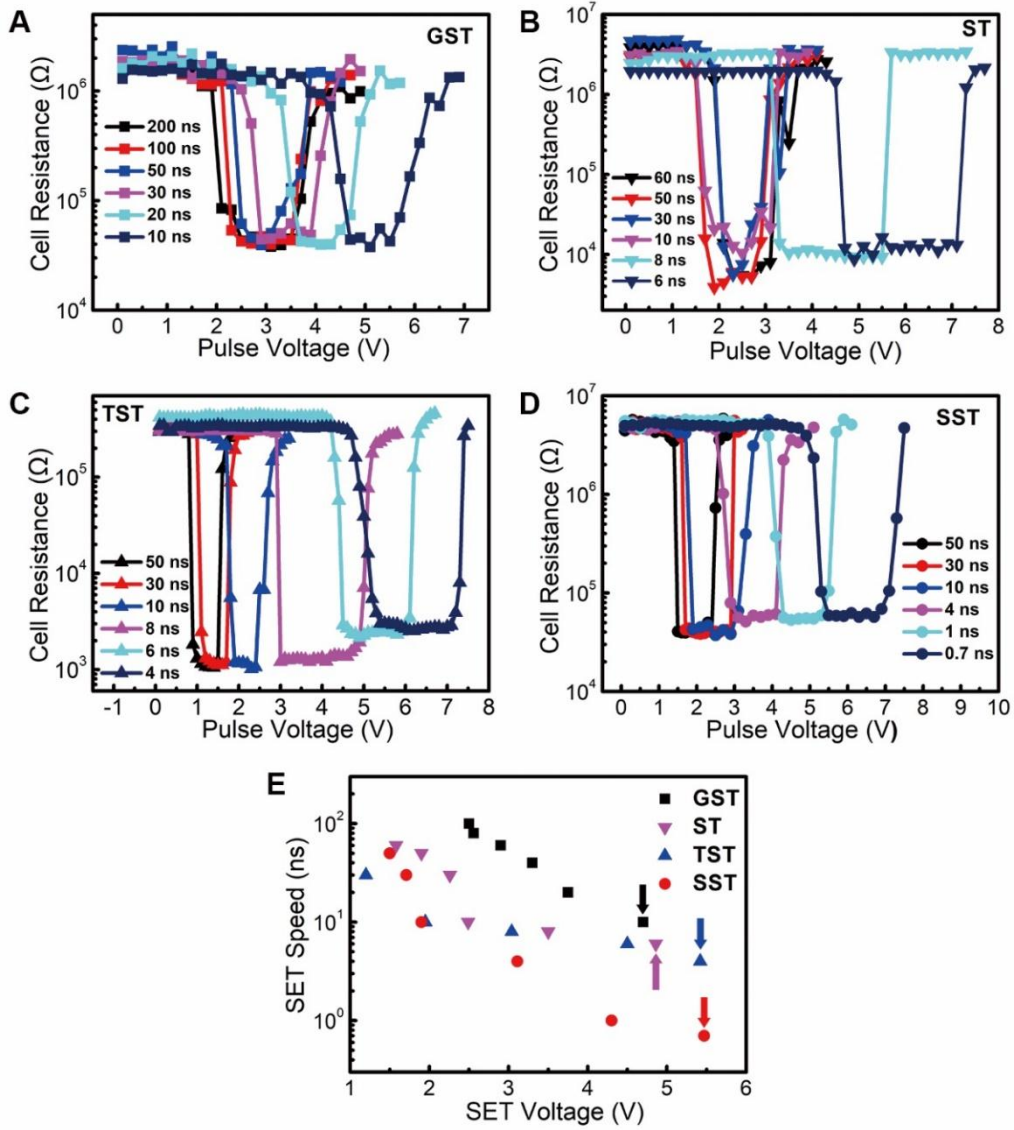


Fig. S6. (A)-(D) The Resistance-voltage characteristics of GST, Sb_2Te_3 (ST), TST, and SST based T-shaped PCRAM devices with the same geometry, e.g., W BEC of 190 nm in diameter. The SET-RESET programming windows are obtained under different pulse widths. As the pulse width decreases, larger input bias is needed to accomplish successful SET and RESET operations. The fastest SET speed for GST, ST, TST, and SST device is 10 ns under ~ 4.7 V bias, 6 ns under ~ 4.7 V bias, 4 ns under ~ 5.3 V bias, and 700 ps under ~ 5.3 V bias, respectively. The time interval between adjacent driving pulses is set to be 2 ms to avoid cumulative effects of multiple-pulse action. (E) The comparison of SET speed for all the devices. Note the vertical axis uses logarithmic scale. The fastest speed for each device is marked with arrow. The SET speed is governed by the completion of the crystallization of the phase change material, and can thus reflect the crystallization speed.

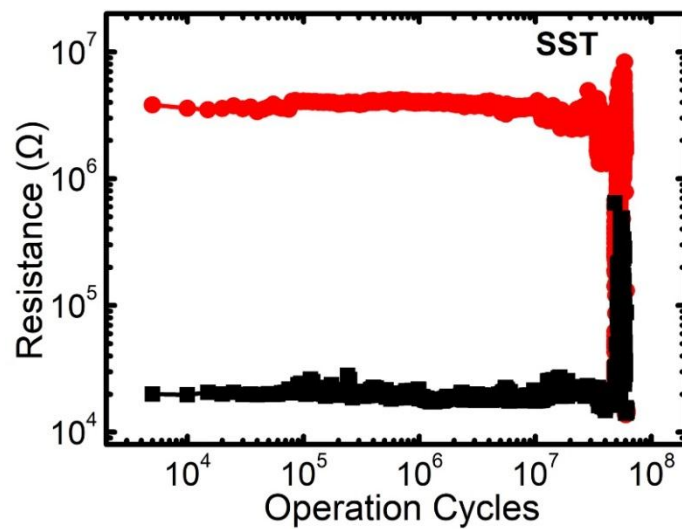


Fig. S7. Up to $\sim 4 \times 10^7$ cyclability of the SST device: SET (@ 1.8 V) and RESET (@ 2.5 V) with 50 ns voltage pulses.

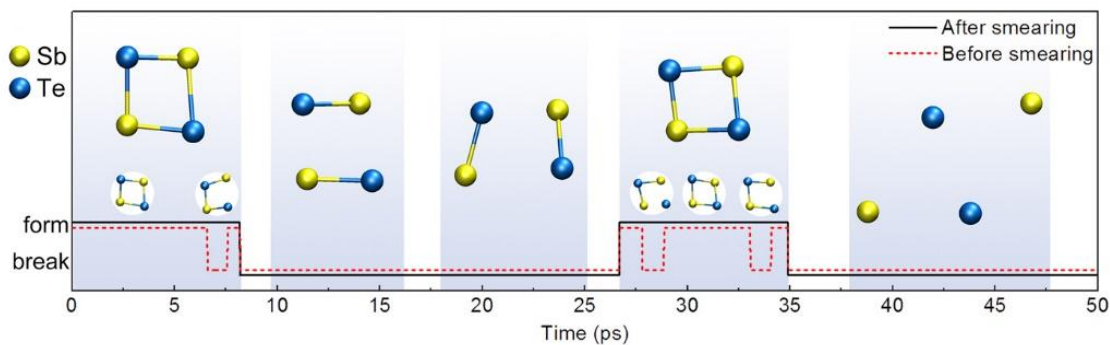


Fig. S8. Sketch for a typical ABAB ring evolving at ~ 600 K. At such high temperature, the ring can break and reform. To calculate the lifetime of the rings, transient fluctuations (seen as short spikes on the red dashed line) were excluded, which are those that last < 1 ps (500 MD steps). These were smeared out for the lifetime calculations (black line). Then the lifetime is calculated by summing up the duration of the “form” periods (two examples in the figure) and divided by the number of these periods. The lifetime is averaged over all the ABAB rings, and over three independent samples. The cutoff for bonds is set as $3.4 \text{ \AA}$.

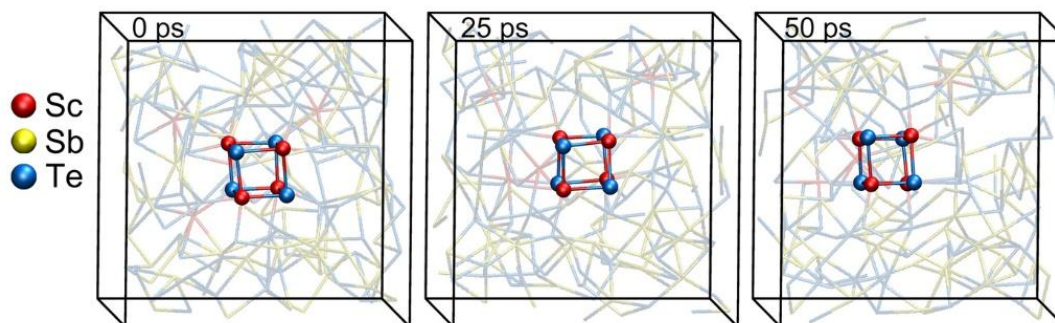


Fig. S9. Another independent SST sample at ~ 600 K. The ScTe cube is pre-fixed during melt-quench, and the constraint is released when the sample is annealed at ~ 600 K. The ScTe cube is robust over 50 ps. This observation is in line with what is presented in Fig. 4B, where the ScTe cube is formed during melt-quench without any constraint. The cutoff for bonds is set as $3.4 \text{ \AA}$.

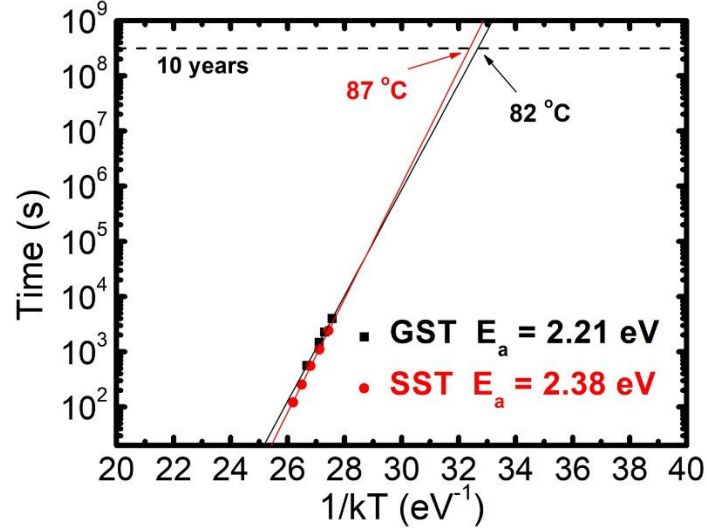


Fig. S10. “Time to failure” versus reciprocal temperature under isothermal heating for both SST and GST devices. The data was fitted using the Arrhenius equation $t = \tau \exp(E_a/k_B T)$, where t is the time to failure when sheet resistance of amorphous film, at a certain isothermal heating temperature, falls to half of its initial magnitude, τ is a proportionality constant, E_a is the activation energy, and k_B is the Boltzmann constant. The good thermal stability of amorphous SST is not in contradiction with the ultrafast crystallization process observed at elevated temperatures, as crystallization is accomplished by firstly nucleation and then crystal growth. Alloying Sc into Sb_2Te_3 increases the viscosity, slowing down diffusion, particularly at low temperatures due to the increased barrier energy and hence temperature dependence. Even if amorphous SST contains crystalline precursors around room temperature, they cannot develop into nuclei and grow, due to the suffocated kinetics.

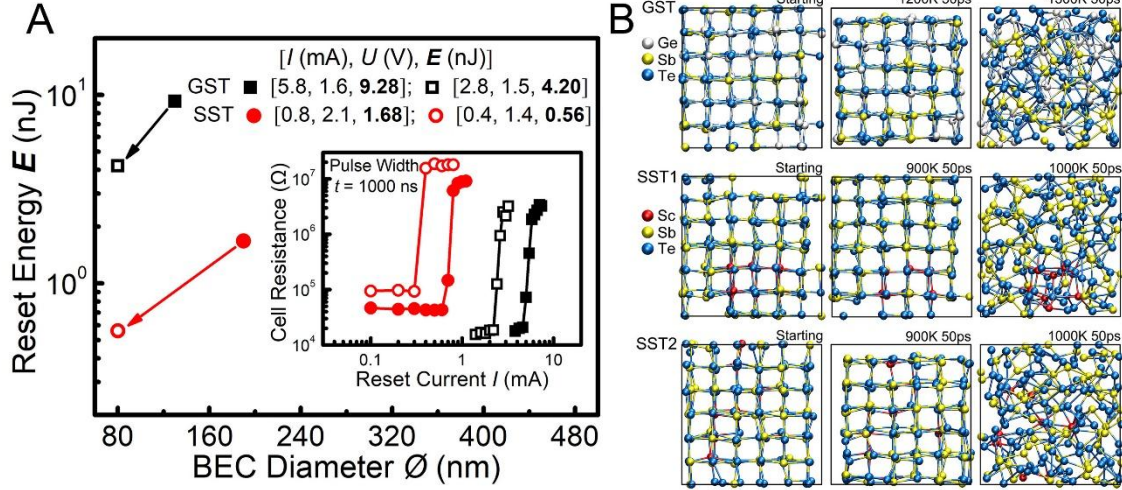


Fig. S11. (A) RESET energy (E) as a function of bottom electrode contact (BEC) diameter (Φ) for the GST and SST devices. Inset shows resistance versus RESET current (I) curves with a fixed pulse width of 1000 ns (t). Transient RESET voltage (U) across the device is recorded once the RESET state is reached. The input RESET energy was calculated as $E = I \times U \times t$. (B) DFMD simulations of rocksalt GST and SST show that GST and SST can keep the rocksalt structure after 50 ps at 1200 K and 900 K, respectively. The former gets disordered within 50 ps at 1300 K, while the latter only needs 1000 K to achieve the same. A possible explanation for this easier amorphization in SST is that the atomic vacancy is richer (1/6) in SST than in GST (1/10), which increases the possibility to accommodate energetically unfavorable Te-Te antibonds, leading to easier disordering. As a result, rocksalt SST consumes less RESET power than rocksalt GST. The disordering process is not very sensitive to the distribution of Sc impurities, as shown by the two sets of SST model (Sc impurities are centered in SST1 but well-separated in SST2), although the in ScTe-segregated model, disordering started in the SbTe region first, before affecting the ScTe region.

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